Conveyance/Transportation of Water

There are two stages in the transportation of water:

- Conveyance of water from the source to the treatment plant.
- Conveyance of treated water from treatment plant to the distribution system.

- In the first stage water is transported by gravity or by pumping or by the combined action of both, depending upon the relative elevations of the treatment plant and the source of supply.
- In the second stage water transmission may be either by pumping into an overhead tank and then supplying by gravity or by pumping directly into the water-main for distribution.



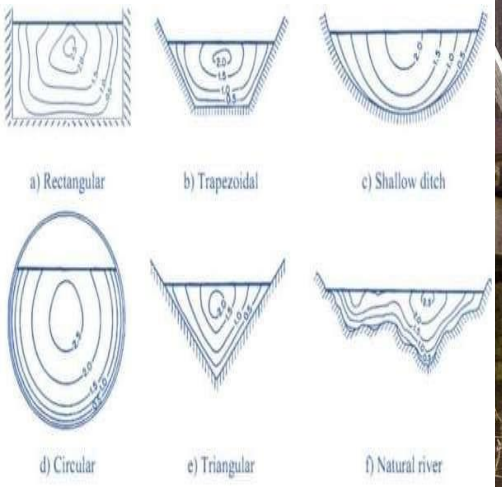
Free Flow (Gravity) Systems vs Pressure Systems

- *In Free-Flow Systems*, water in the conveying section flows freely due to gravity. The hydraulic gradient line coincide with the water surface. Very long conveying sections may be needed to suit the slope of the existing ground. Typical sections used for free-flow are: canals, flumes, grade aqueducts and grade tunnels.

- *Pressure Systems* are closed conduits where the water flows under pressure above the atmospheric pressure. These are independent of the natural hydraulic gradient line and thus can follow the natural ground surface, requiring lesser length of conduit. The pressure conveyance systems are mostly in the form of pipes (which could be of various types depending on construction material), however closed pressure aqueducts or pressure tunnels are also used, but not common.

Conveyance of Water: Open Channels and Flumes

- Occasionally used to convey the water from the source to the treatment plant.
- Ground channels are easily and cheaply constructed by cutting in high grounds and banking in low grounds.
- Flumes may be elevated, supported above the ground over trestles etc, and may be constructed with R.C.C, wood or metal.
- The channels should be lined properly to prevent the seepage and contamination of water.
- A uniform longitudinal slope is provided to ensure gravity flow, however the hydraulic gradient line is controlled within allowed limits to avoid scouring.
- Water loss by evaporation, and seepage (if unlined) are major concerns for open channels.

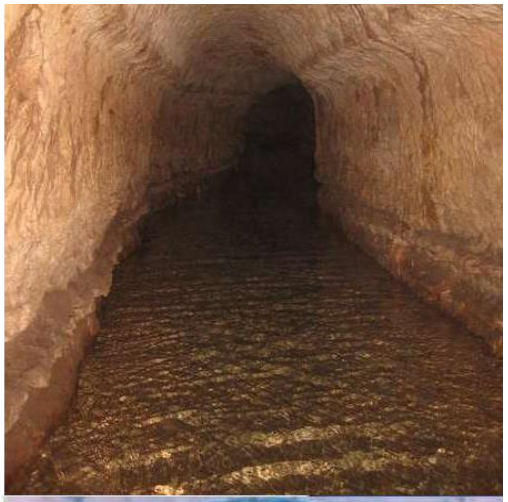






Conveyance of Water: Tunnels and Aqueducts

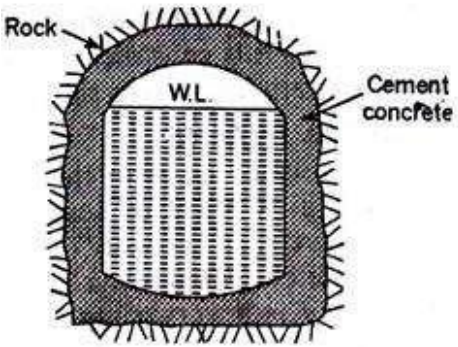
- Aqueducts and Tunnels the closed conduits constructed with masonry, cement concrete or RCC and used for conveying water from source to the treatment plant or point of distribution.



- These are mostly designed as gravity conduit, but sometimes pressure aqueducts or pressure tunnels are also used for conveying water.



- These are mostly horse-shoe or rectangular cross-section, but pressure tunnels have circular cross-section. These are generally water tight leading to very little loss of water.



- The average velocity is usually kept about 1 m/sec

Conveyance of Water: Pipes

- Pipes are circular conduits used for water flows under pressure or gravity.
- Pressure pipes are most common these days and have nearly eliminated the use of channels, aqueducts and tunnels to a large extent.



- These are made of various materials like:
  - Cast Iron (CI) Pipes
  - Steel Pipes
  - Galvanized Iron (GI)
  - Pipes Copper Pipes
  - Plastic or Polythene or PVC pipes
  - Asbestos Cement (AC)
  - Concrete Pipes



Type of Pipe Material for Pipelines for Water Distribution

Various types of pipes are used for water supply system including metallic and non-metallic pipes. Most common types of pipes used for water supply system are:

a. Galvanised Iron Pipes – metal pipe

b. Mild Steel Pipes metal pipe

c. Poly Vinyl Chloride pipes - non- metal pipe

d. High Density Poly Ethylene Pipes - non metal pipe

e. Ductile Iron Pipes

GALVANISED IRON (G. I . PIPES)







Galvanised Iron (GI) Pipes

- Cheap in cost and light in weight.
- Light in weight and easy to join.
- Affected by acidic or alkaline water.
- GI pipes are highly suitable for distribution

system. They are available in light (yellow colour code), medium (blue colour code) and heavy grades (red colour code) depending on the

thickness of pipe used.

Normally, medium grade pipes (wall thickness 2.6-4.8 mm) are used for water supply system. Normally, 15-150 mm size pipes (nominal internal diameter) are used for distribution system. They are available in length of 3 m.

- GI pipes can be used in non-corrosive water with pH value greater than 6.5.
- GI pipes can be used for rising main as well as distribution.
- GI pipes are normally joined with lead putty on threaded end.

Mild Steel Pipes



- Number of joints are less as they are available in longer length.
- Pipes are durable and can resist high internal water pressure and highly suitable for long distance high pressure piping.
- Flexible to lay in certain curves.
- Light weight and easy to transport. Damage in transportation is minimal.
- Pipes are prone to rust and require higher maintenance.
- Require more time for repairs and not very suitable for distribution piping.
- Available in diameter of 150-250 mm for water supply and cut lengths of 4 - 7 m (2.6-4.5 mm wall thickness).
- Steel Pipes are joined with flanged joints or welding.

Poly Vinyl Chloride (PVC unplasticised) Pipes

- Cheap in cost and light in weight.
- Economical in laying and jointing.
- They are rigid pipes.
- Highly durable and suitable for distribution network..
- Free from corrosion and tough against chemical attack.
- Good electric insulation.
- Highly suitable for distribution piping and branch pipes.
- Less resistance to heat and direct exposure to sun. Hence, not very suitable for piping above the ground.
- PVC pipes weigh only 1/5th of steel pipes of same diameter.
- Certain types of low quality plastic impart taste to water.
- Available in size 20-315 mm (nominal internal diameter) for water supply with pressure class of 2.5, 4, 6, 8 & 10 kg/cm2 for water supply. Ideally pipes with 6 kg/cm2 should be used.
- Classification of pipes is done according to its pressure class.

PVC PIPES





HDPE(HIGH DENSITY POLYETHYLENE)

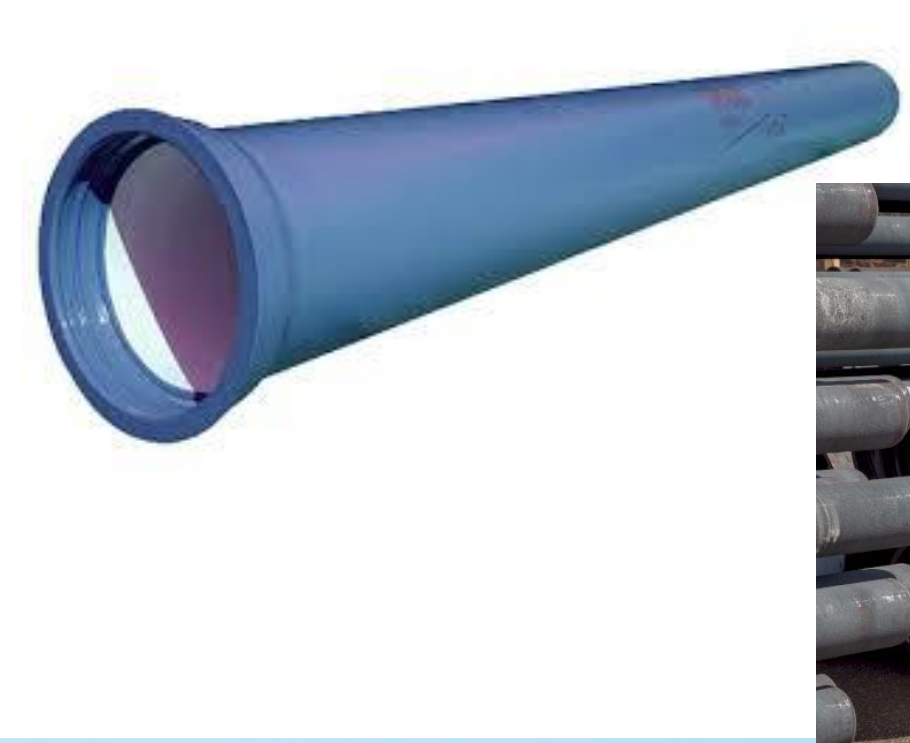
- Light in weight.
- Flexible than PVC pipes.
- HDPE pipes are black in colour.
- Suitable for underground piping and can withstand movement of heavy traffic.
- Allows free flowing of water.
- Highly durable and suitable for distribution network.
- Free from corrosion..
- Good electric insulation.
- Useful for water conveyance as they do not constitute toxic hazard and does not support microbial growth





Ductile Iron Pipes

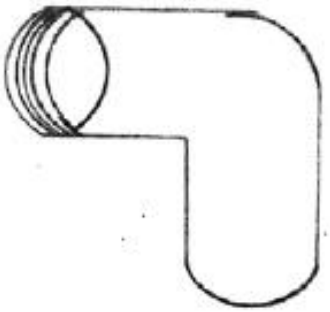
- Ductile Iron pipes are better version of cast iron pipes with better tensile strength.
- DI pipes are prepared using centrifugal cast process.
- DI pipes have high impact resistance, high wear and tear resistance, high tensile strength, ductility and good internal and external corrosion resistance.
- DI pipes are provided with cement mortar lining on inside surface which provides smooth surface and is suitable for providing chemical and physical barriers to water. Such pipes reduce water contamination.
- The outer coating of such pipes is done with bituminous or Zinc paint.
- DI pressure pipes are available in range from 80-1000 mm diameter in lengths from 5.5-6 m.
- Available in thickness class K7 and K9 with barrel wall thickness ranging from 5-13.5 mm. Also available in pressure class (Like C25, C30, C40 etc.).
- They are about 30 percent lighter than conventional cast iron pipes.
- DI pipes lower pumping cost due to lower frictional resistance



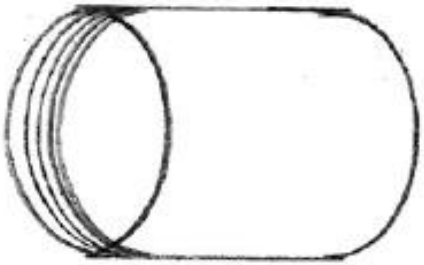



Kubota JAPAN

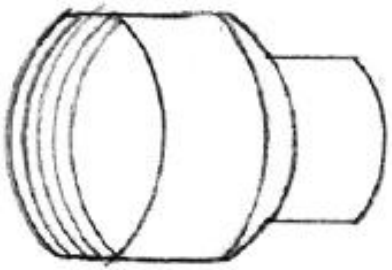
Type of Pipe Fittings



Elbow



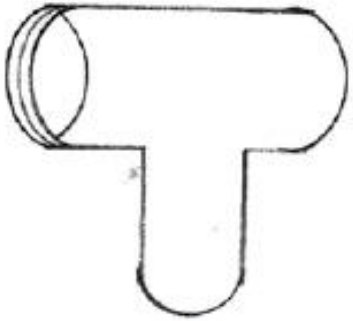
Coupling



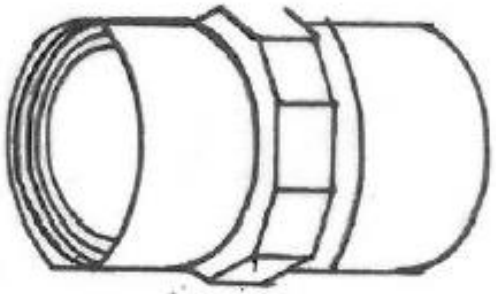
Reducer



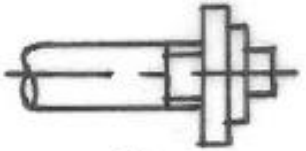
Bend



Tee



Union



Plug



Nipple



Plain End Bend



Single Flange Bend



Double Flange Bend



Short Pipe



Plain End Tee



Off Take Tee



All Flange Tee



M. S Pipe



Plain End Taper



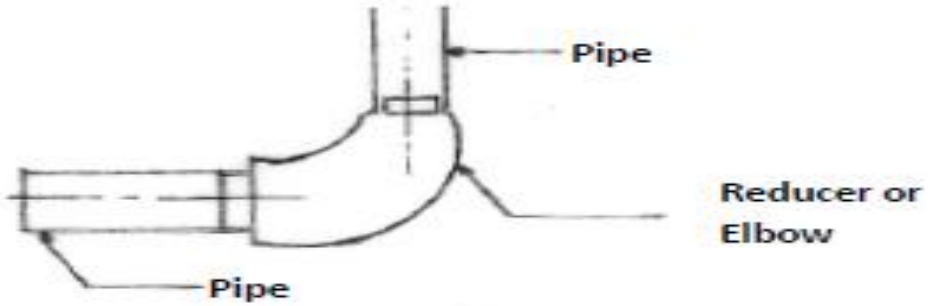
Single Flange Taper

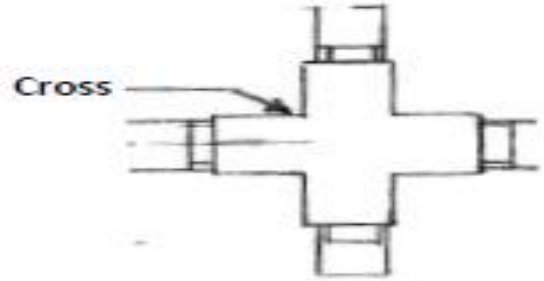


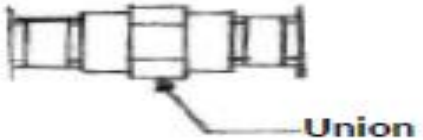
Double Flange Taper

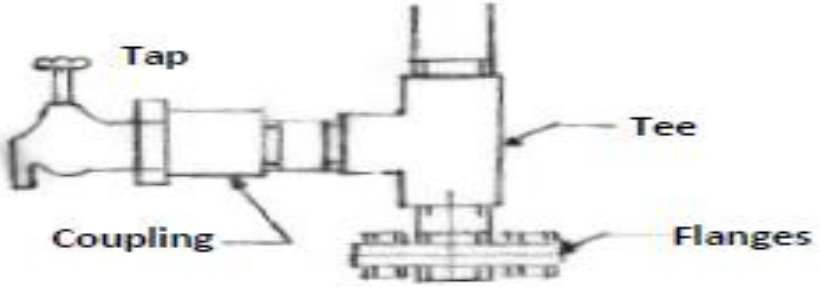


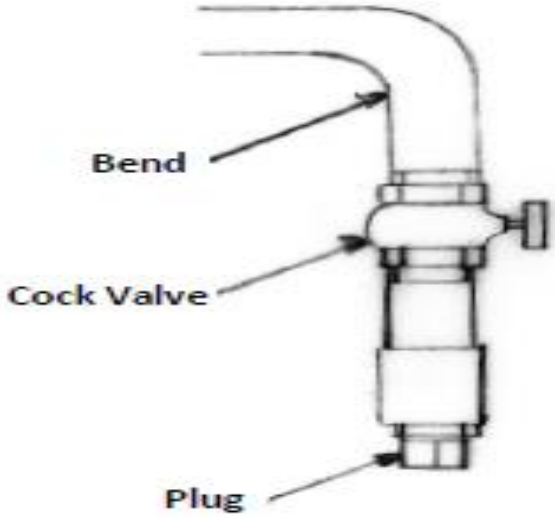
Welding Collar











Typical piping Layout with pipe fittings

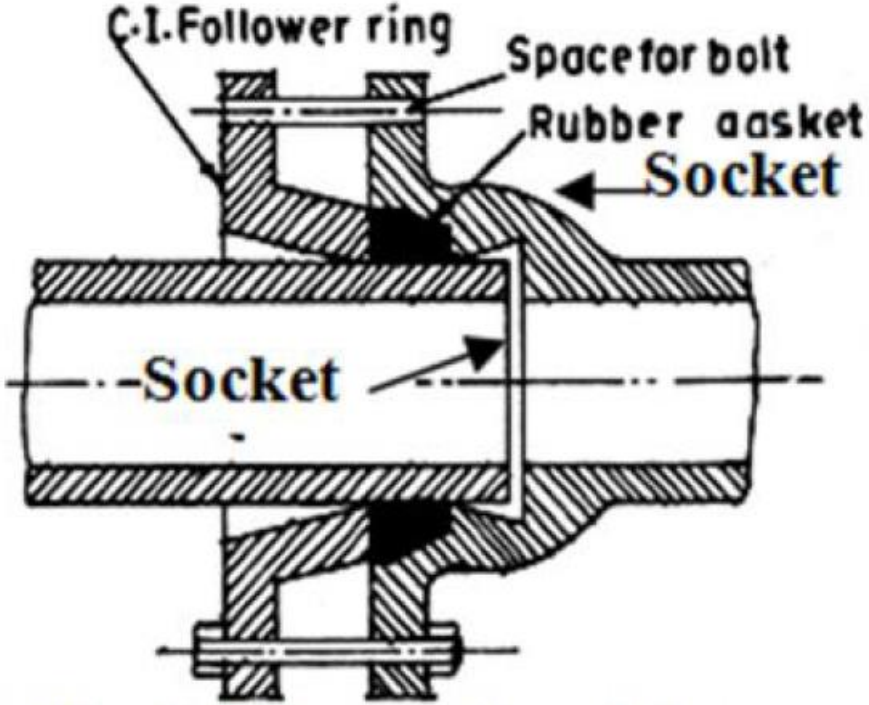


Fig 8.3 Expansion joint

(Source: Birdie and Birdie, 2001)

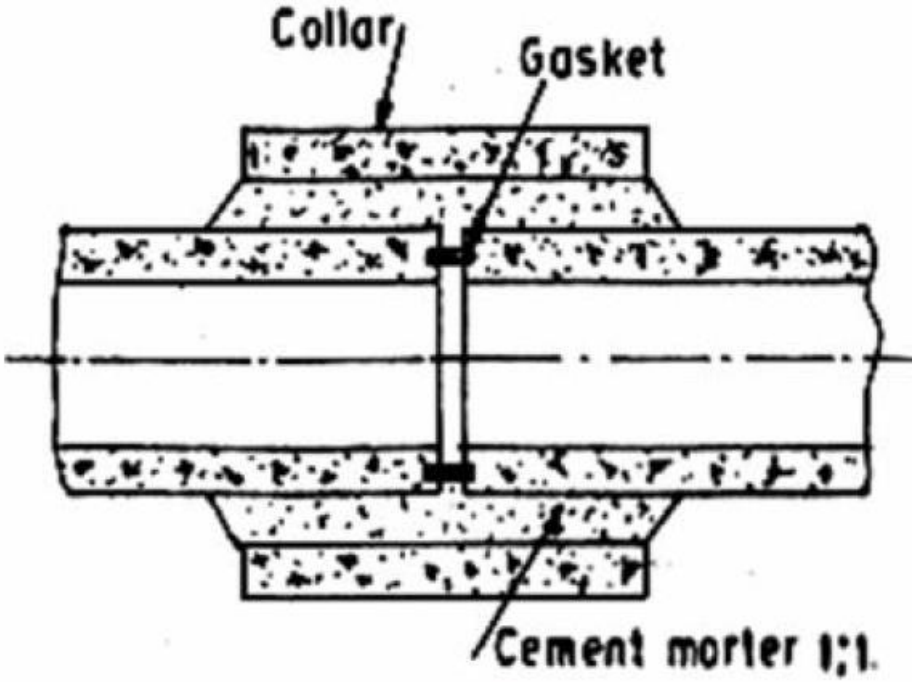


Fig 8.4 Collar joint

(Source: Birdie and Birdie, 2001)

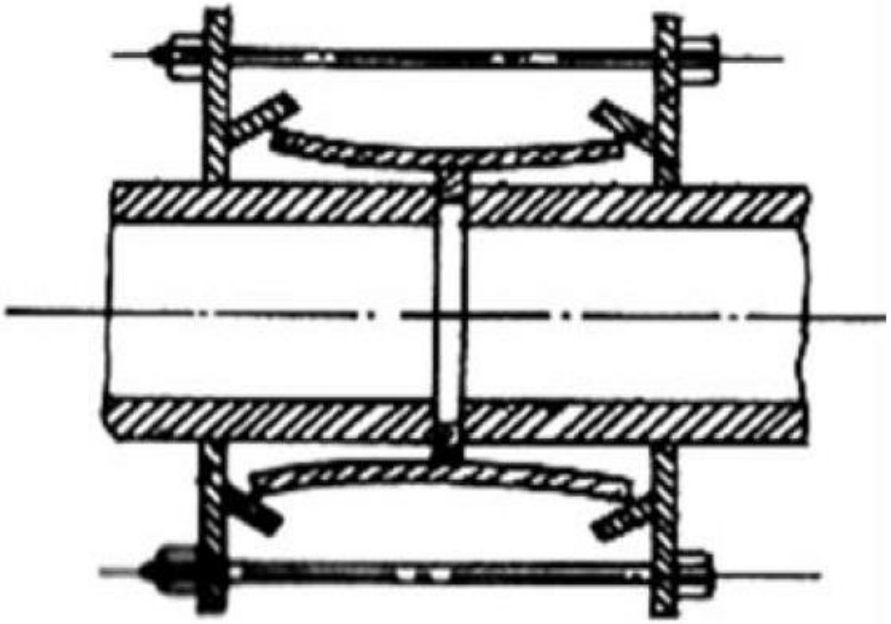


Fig 8.6 Mechanical joint
(Source: Birdie and Birdie, 2001)

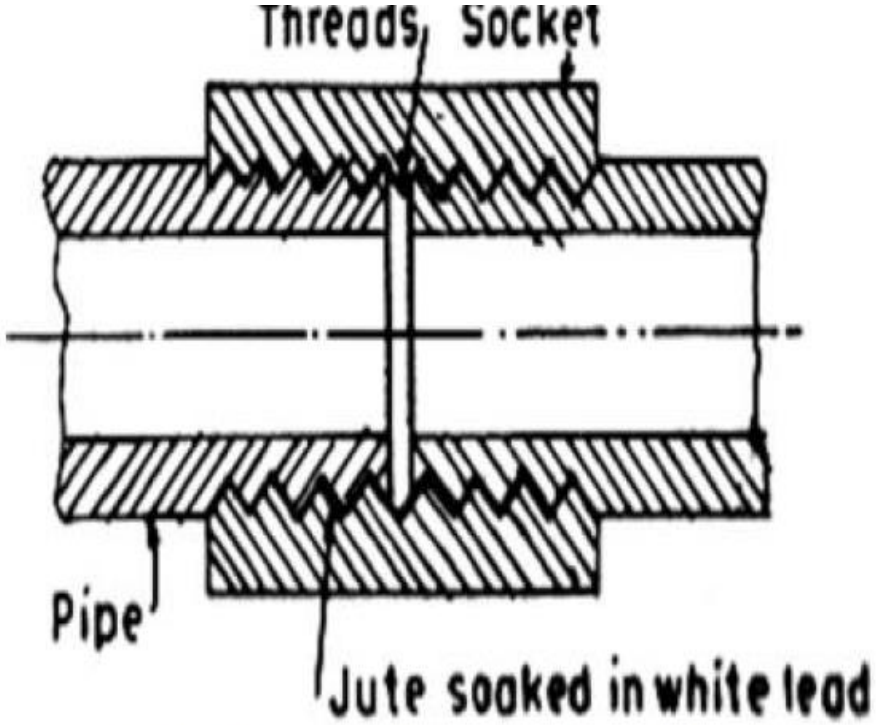


Fig 8.5 Screwed & socket joint
(Source: Birdie and Birdie, 2001)

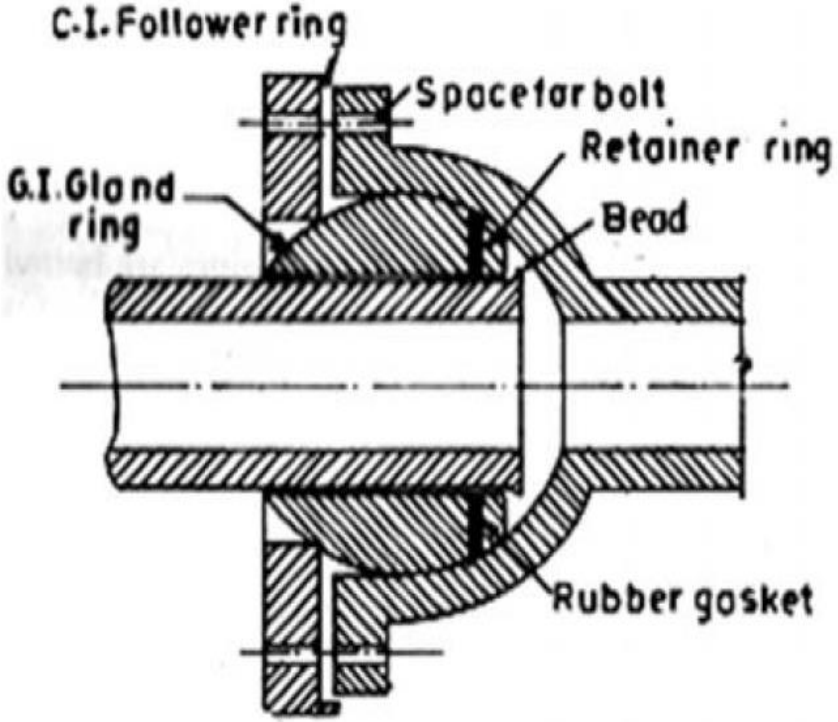


Fig 8.7 Flexible joint

(Source: Birdie and Birdie, 2001)

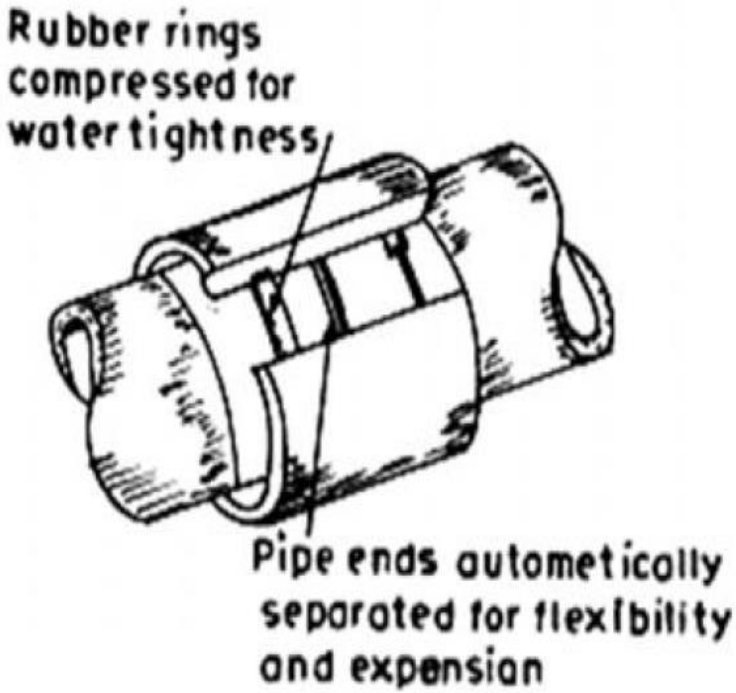
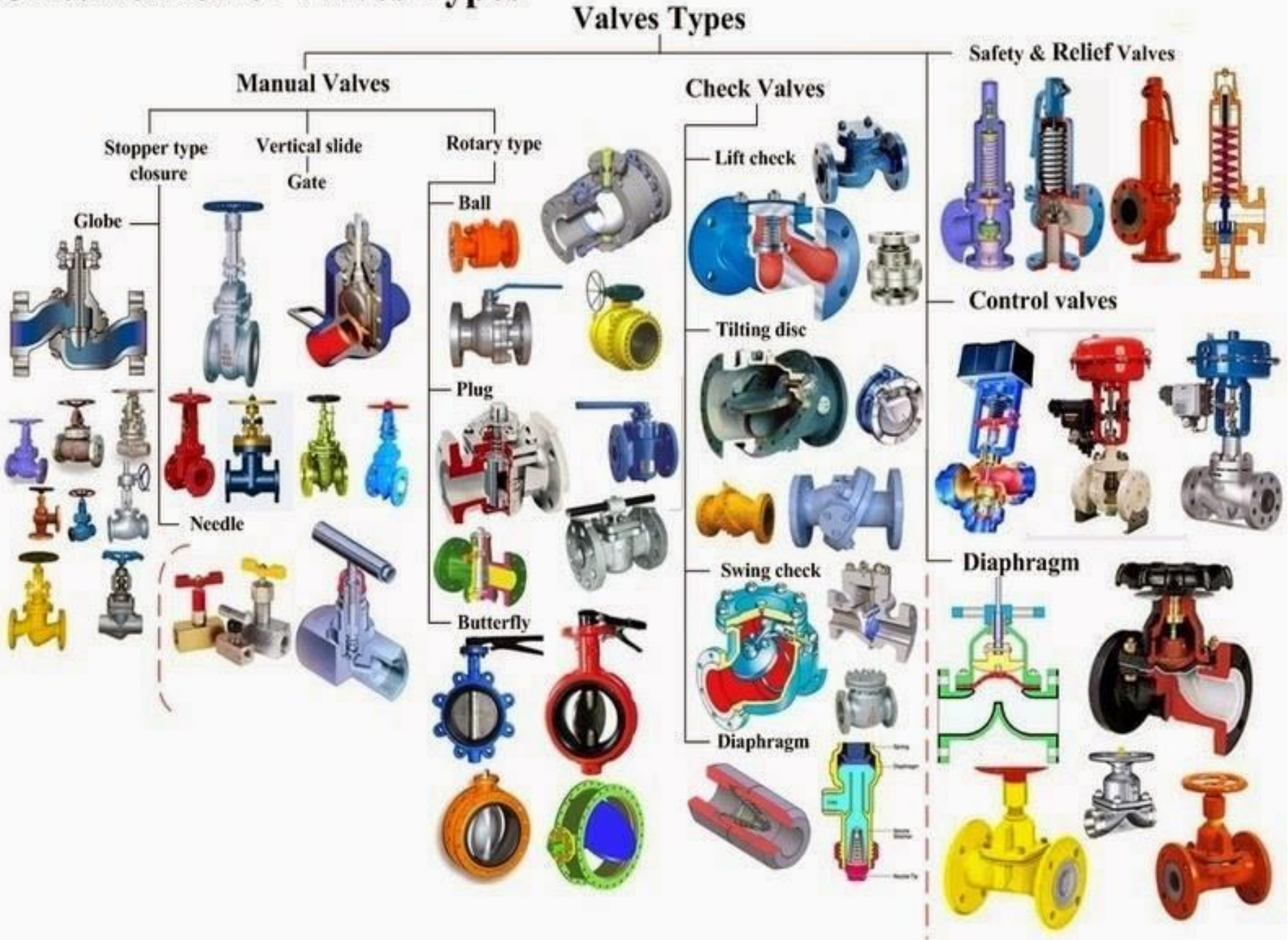


Fig 8.8 AC pipe joint

(Source: Birdie and Birdie, 2001)

Classification of Valves Types



Gate Valves

- The Gate valves are characterized by a siding gate which is moved by the stem perpendicular to the direction flow.

Application –

- It is used for on-off application. Suited for high temperature and pressure use with variety of fluids.
- They are not primarily used for slurries, viscous fluid etc

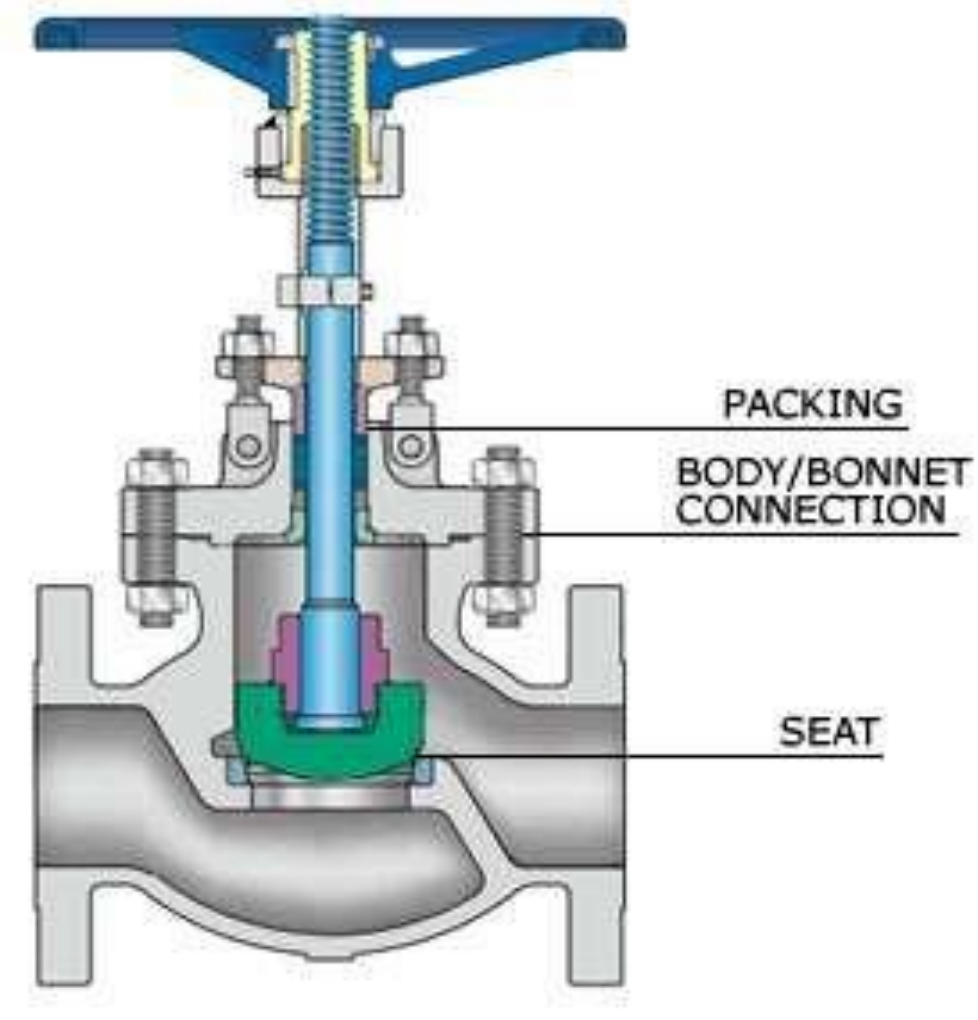
Advantages of Gate Valve –

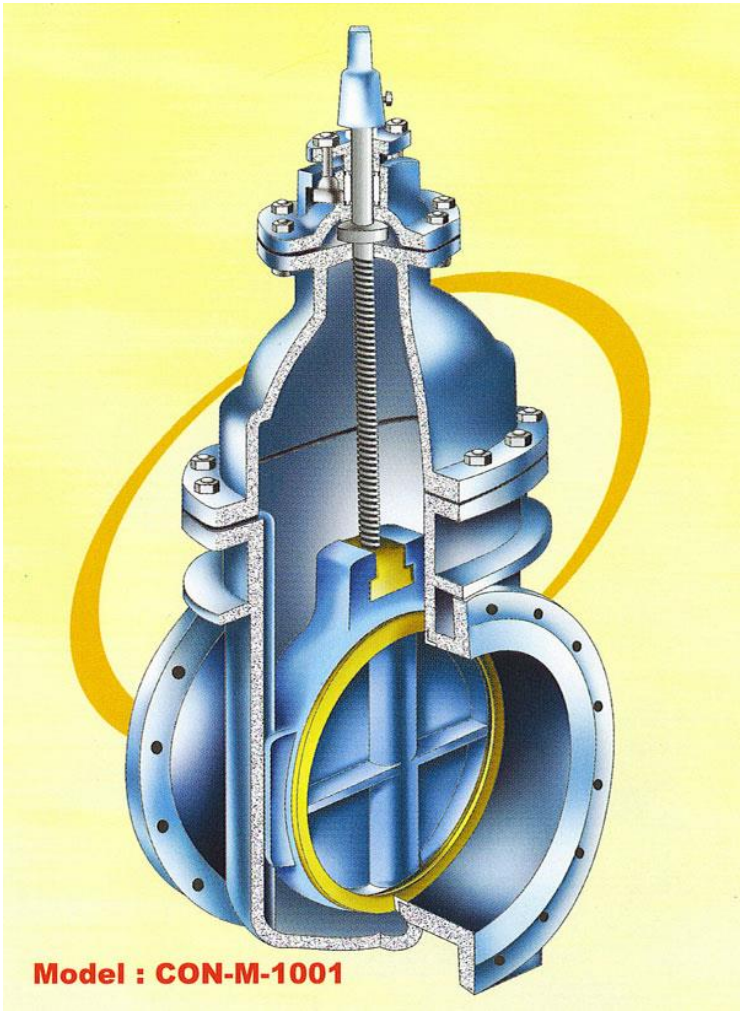
Low pressure drop when fully open and tight sealing

Disadvantages of Gate Valve –

- Causes vibration, seat disc wear in partial open condition. Slow response characteristics and require large actuating force.

GATE VALVE





Globe Valve

In globe valves disc or plug is moved on or off the seat. The seat opening is directly proportional to the travel of the plug. It is short stem travel, high seating capacity, large pressure drop and high flow controllability.

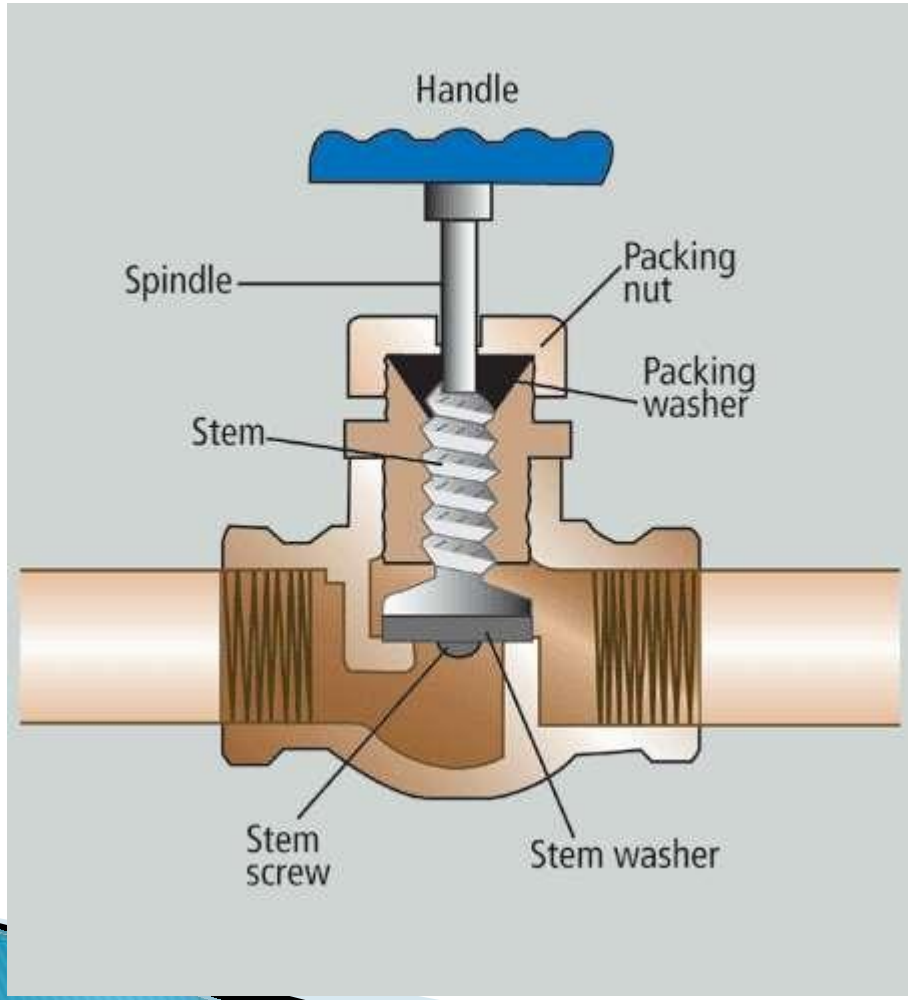
Application –

It is used primarily for throttling purposes. It may be considered a general purpose flow control valve high temp application.

Advantages of Globe Valve –

- Faster to open or close, most reliable form of seating,throttling to control the flow to any desired degree, positive shut-off.
- **Disadvantages of Globe Valve** - Seat disc wear in partial open conditions.

GLOBE VALVE


Handle
Spindle
Packing nut
Packing washer
Stem
Stem screw
Stem washer


6
150
WCB



SLUICE VALVE



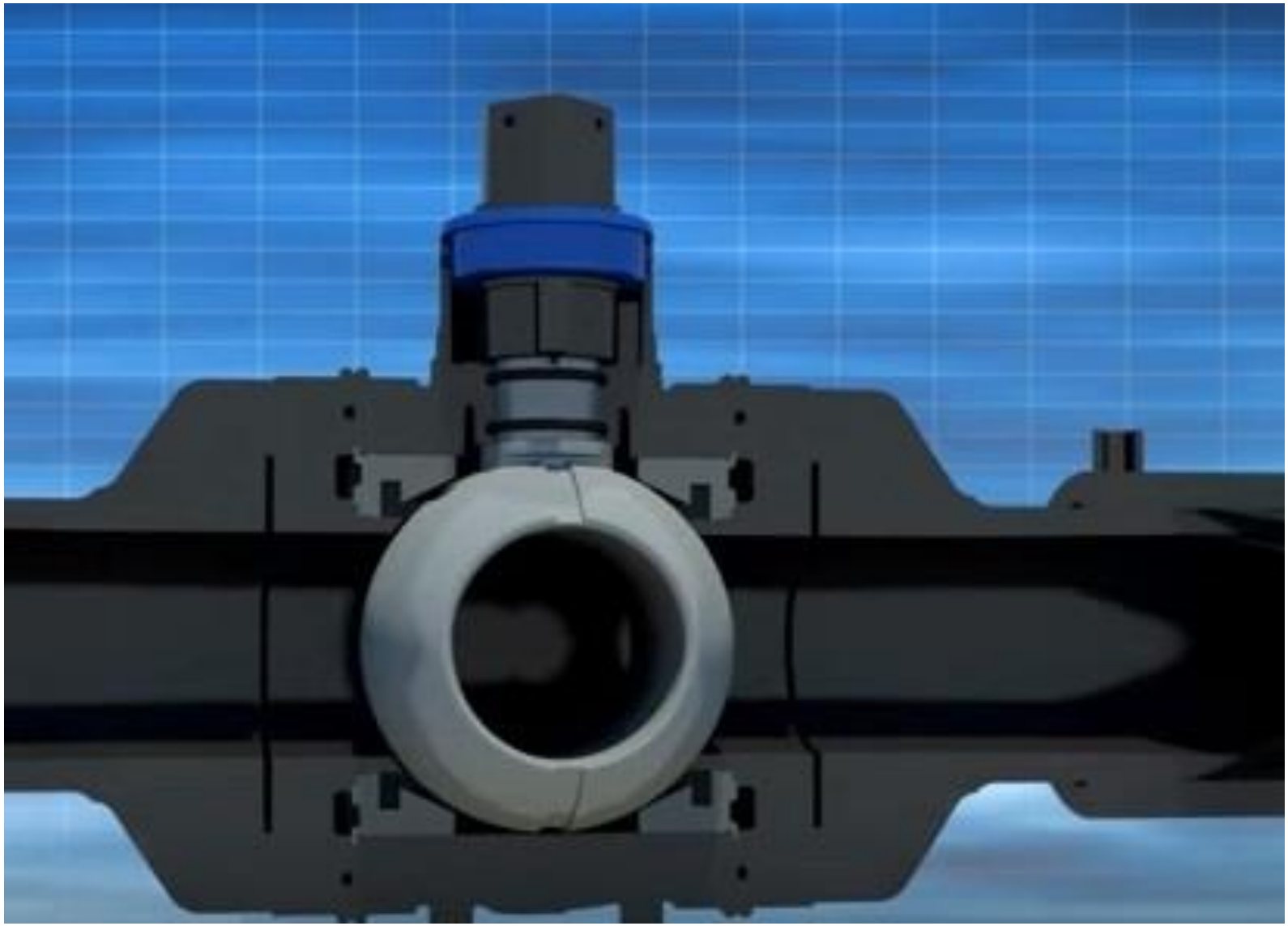
SCOUR VALVE

- **Plug Valves**
- It consist of body, plug and cover. The plug is tapered or cylindrical. IN the open position, the bore in the plug connects the inlet and outlet ends of the valve providing straight line flow.
- Application - The plug valves are extensively used in refinery,petrochem and chemical industries and for general purpose involving on off services.
- Advantages of plug valve - Normally small in size, require less headroom and available in wide range of materials. They provide tight shut off, quick opening and low pressure drop.
- Disadvantages of plug valve- Plug valve may be subjected to galling



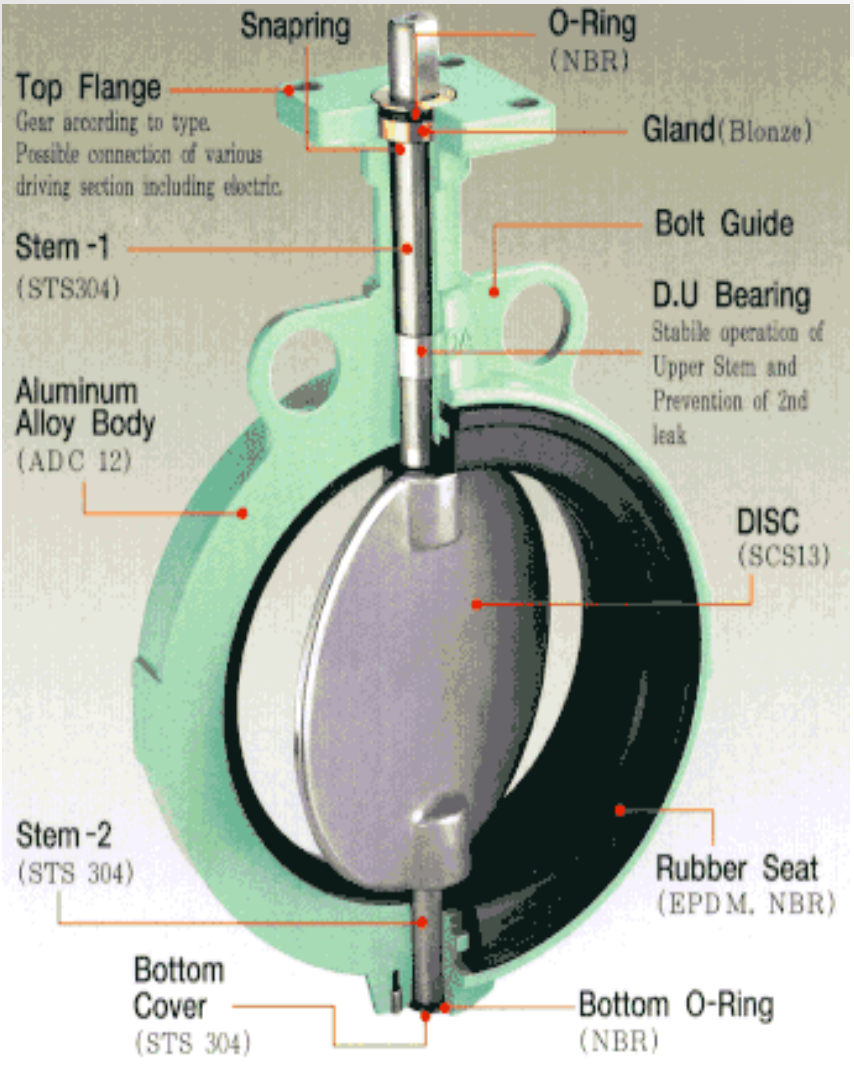
Ball Valves

- It is improvisation of the plug valve. It is basically a ported sphere in a housing. The seat matching the ball is circular so that the seating stress is circumferentially uniform.The seats are usually made up of PTFE which is inert to all chemicals, has low coefficient of friction and resiliency.
- Application - It is used for flow and pressure control and shut off and for corrosive fluids,slurries,normal liquid and gases. For high temp and pressure.
- Advantages of Ball Valve - Low pressure drop shut-off,quarter turn operation, easy to maintain, low torque. They are small in size and low in weight.
- Disadvantages of Ball valve - PTFE seats are subjected to extrusion is the valve is used for throttling. Fluid trapped in the ball in the closed position may cause problem of build up of vapor pressure and corrosion.



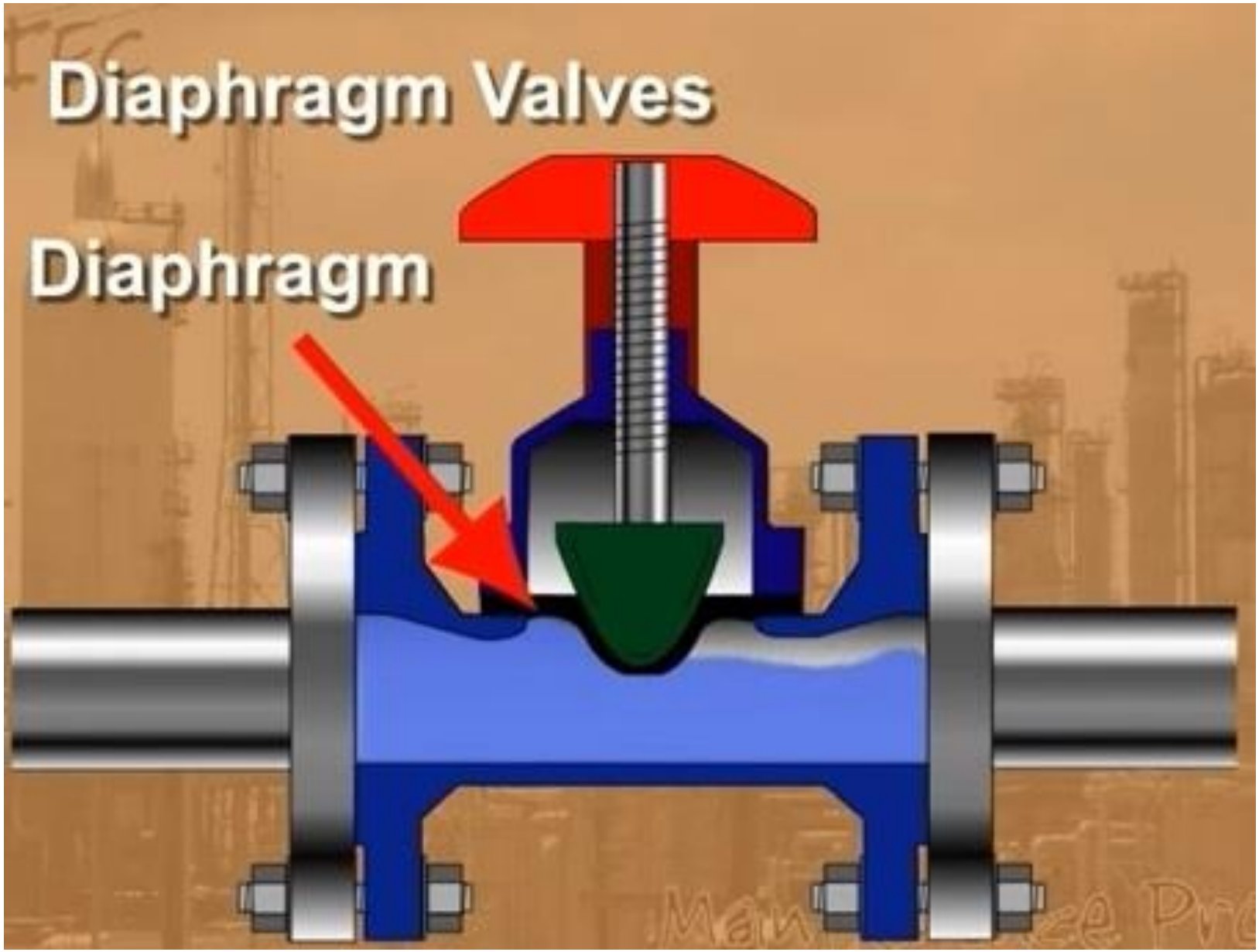
- **Butterfly Valves**
- They are rotary valves with a disc rotating at right angles within a pipe section body.
- The stem passes through the disc is supported by both ends of the body.
- The shaft is ensured to the disc either by bolts or pins. Rotating stem through 90 degrees. Fully opens or closes the valves.
- Application - Large flow control, Low pressure applications where leakage is unimportant.
- Advantages of Butterfly Valve - Simple, Compact form, Quick opening, Good control, large flow, low pressure drop, and low weights and costs.
- Disadvantages of Butterfly Valve- No tight shut off

Butterfly Valve





- **Diaphragm Valves**
- The flexible diaphragm is connected to a support member known as compressor, which represents the closure member. When the valve is opened the diaphragm is lifted out of the flow path and the fluid has a smooth streamlined flow. When the valve is closed the diaphragm is tightly seated against the seat in the body known as the weir. The diaphragm is of NEOPRENE.
- Application - Used for corrosive liquids at low temp and pressure.
- Diaphragm Valves Advantages - The diaphragm completely keeps the working parts in isolation from the process fluids. Simple in construction, easier to operate and maintain.
- Disadvantages of Diaphragm valves- Not for high temp and pressure.


Diaphragm Valves
Diaphragm

TAP OR BIB COCK













STOP COCK





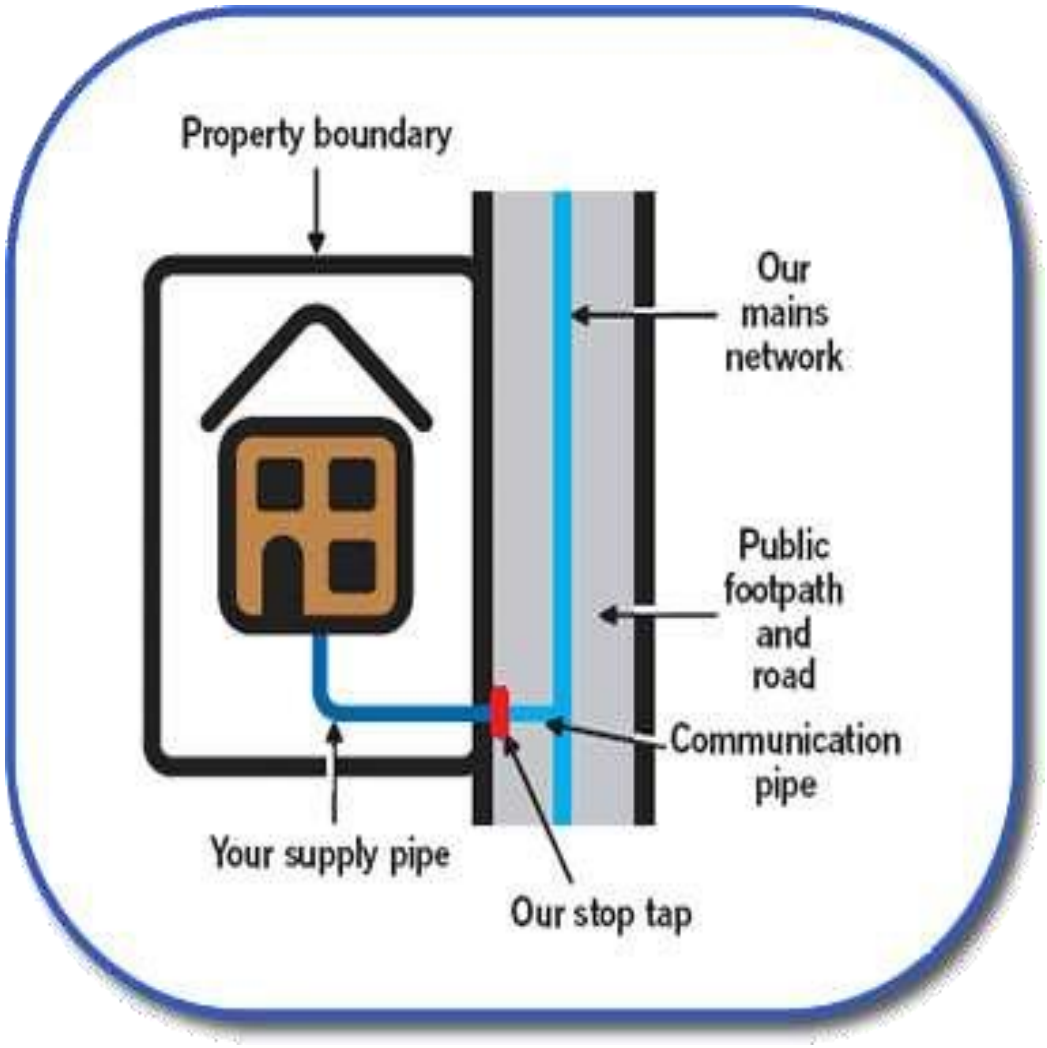
VARIOUS FITTINGS –GLOBE VALVE





STOP TAP




Property boundary
Our mains network
Public footpath and road
Communication pipe
Your supply pipe
Our stop tap

VARIOUS VALVES , TUBES AND PIPES



Copper Pipe Clips 22mm



Copper Tube 15mm × 20m



Micro bore Copper Tube 10mm × 25m



Pegler Ball Valve Red 15mm



Pegler Double Check Valve 15mm



Pegler Gate Valve 22mm



Pegler Washing Machine Tap 15mm x ¾""



PTFE Tape 12m Pk of 10

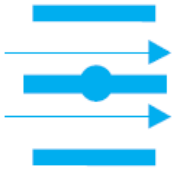
ELECTRICAL MOTORS







Butterfly valve



Open



Closed

- Valve shaped like a butterfly.
- Tight shut-off and can be used as a control valve.
- Little resistance to flow (allows smooth flow).
- Optimal for automated operation with a low operating torque and 90 degrees operating angle.
- Lightweight and compact (large diameter models are also available).

Check valve



Open



Closed

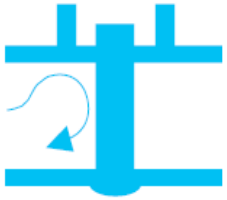
- For use when flow is only in one direction.
- Lightweight disc allows vertical installation.
- High operating speed prevents water hammer.

InstrumentationTools.com

Gate valve



Open



Closed

- Like its name implies, the gate is lowered to cut off the path of flow.
- For use as an on/off valve (not suitable as a control valve).
- Little resistance to flow when fully open (allows smooth flow).
- Long stroke requires time to open and close; not suitable for quick operation.

Globe valve



Open



Closed

- The globe-shaped body controls the fluid into a S-shaped flow.
- Tight shut-off and can be used as a control valve.
- Large resistance to flow (does not allow smooth flow).
- Much power is required to open and close the valve (not suitable for large sizes).

Ball valve



Open



Closed

- Valve stopper is ball-shaped.
- For use as an on/off valve (not suitable as a control valve).
- Little resistance to flow when fully open (allows smooth flow).
- Optimal for automated operation with a 90 degree operating angle.
- Advanced technology is required to manufacture ball.

Item	Gate Valve	Globe Valve	Ball Valve	Butterfly Valve
Disc actuation	Vertical	Vertical	Rotary	Rotary
Manual mode of operation	Round handle Gear operated	Round handle Gear operated	Lever handle Round handle Gear operated	Lever handle Gear operated
Automatic mode of operation	Pneumatic/ Hydraulic Electric actuator	Pneumatic/ Hydraulic Electric actuator	Pneumatic/ Hydraulic Electric actuator	Pneumatic/ Hydraulic Electric actuator
Application	On - off	On - off control	Automatic on - off Can be controlled	Automatic on - off Can be controlled
Operating temperature range	Wide	Wide	Narrow	Narrower
Operating pressure range	Wide	Wide	Wide	Narrower
Face-to-face dimension	Long	Longest	Long	Short
Weight	Heavy	Heavy	Heavy	Light

Rising Mains (Raw Water Mains)

- Rising main is the pipe through which the pumped water is transported to the next unit (generally treatment plant).
- Generally the rising main should convey the water direct from the pump to the high-level storage tank without being tapped or having water drawn off in any way.
- For a pre-fixed demand (discharge), water may be pumped through a bigger diameter pipe at low velocity, or through a lesser diameter pipe at high velocity.
- The larger dia pipes are costlier thus capital cost of the project will increase, however in. Smaller dia pipes are relatively cheaper, but requires operation at higher velocity leading to higher head loss and thus require more power for pumping (increased operation cost).



Economic Diameter of Rising Mains

- The pipe diameter and material is selected such that overall cost of the project should be minimized from both, operation-maintenance and construction point of view.
- The diameter which provide such optimum condition (overall most economical in initial cost as well as maintenance cost) is known as *"economic diameter"* of the pipe.
- *Lea's empirical formula* for economic diameter suggests

$D = 0.97 \text{ to } 1.22 \sqrt{Q}$

Where, D is Economic dia of pipe, in m,

Q= Discharge, in m^3/s

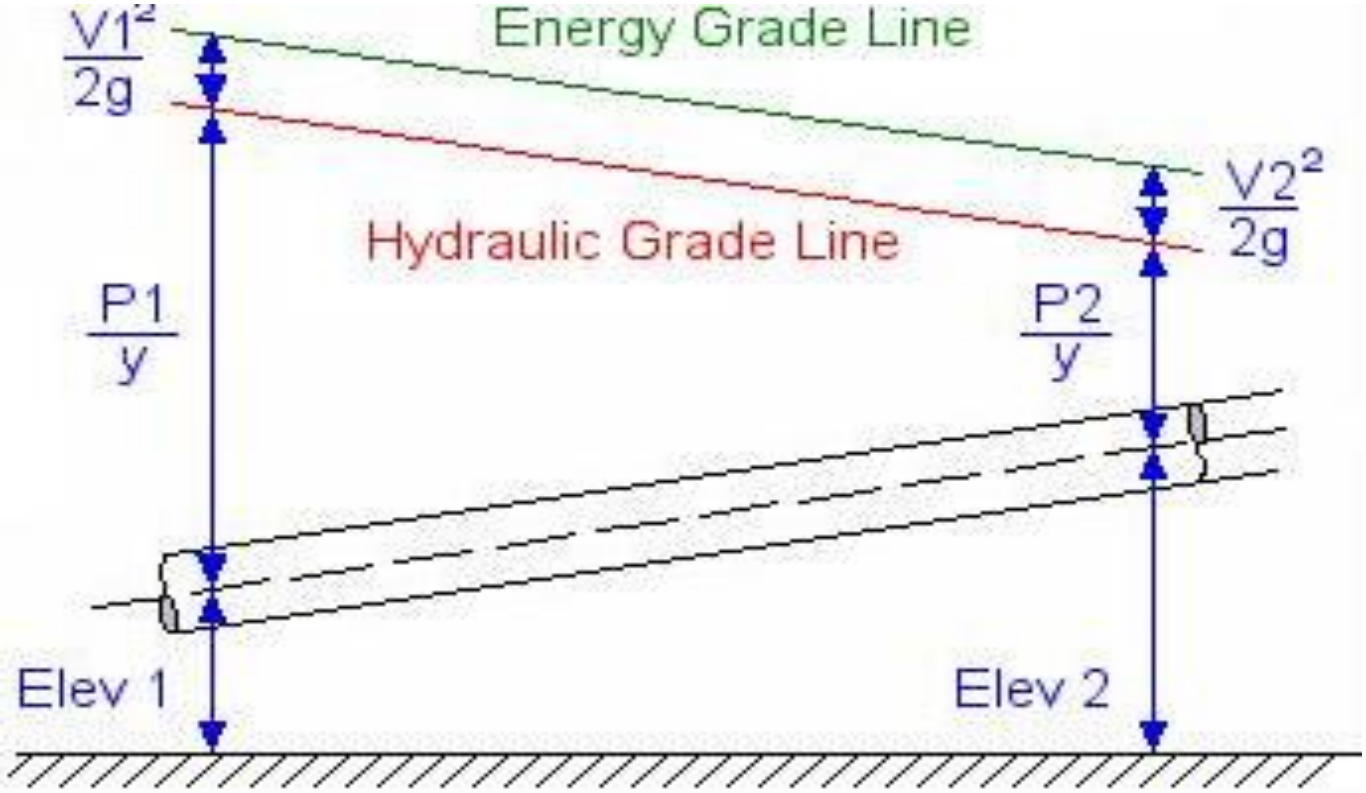
- Generally, a velocity range of 1 – 1.5 m/s is preferred for Rising Mains





HYDRAULICS OF FLOW AND DESIGN OF PRESSURE PIPES

- The pressure pipes can be laid at any depth below the hydraulic gradient line.(for economical consideration; they should follow the HGL)
- Hydraulic gradient line is basically defined as the line which will give the sum of pressure head and datum head or potential head of a fluid flowing through a pipe with respect to some reference line.
- If the pipe goes much below the hydraulic gradient line, the pressure developed in the pipe may be too high for ordinary pipe material.
- It will demand the construction of thicker and stronger pipes, which may create enormous constructional difficulties and increased cost.



- The hydraulic gradient should be such as to generate velocities which are neither so small as to require large dia pipe, nor so large as to excessive loss of pressure head.

- The velocities should also be non-silting and non-scouring.

- The flow velocities are normally kept between 0.9 m/s to 1.5 m/s though velocities upto 3 m/s to 6 m/s can be resisted by the commonly available pipes or pipe materials.

- Design should be in such a way that the available pressure head between the source and the city is just lost in overcoming the frictional resistance offered to the flow by the pipe interior.

- In addition to the head lost due to pipe friction(H_L), there are minor losses caused by the abrupt changes in the flow geometry due to
  - Changes in pipe size
  - Bends
  - Valves and fittings
- In long pipes, the minor losses can be neglected.
- The head loss due to pipe friction can be found by using
  - Darcy-Weisbach formula
  - Manning's formula
  - Hazen-William's formula and
  - Modified Hazen-Willaim's formula

Hydraulic Design of Conduits

- **The design of water supply conduits depends on the resistance to flow, available pressure or head, and allowable velocities of flow.**
- **Although there are several formulae, generally, Hazen-William's formula for pressure conduits and Manning's formula for free flow conduits are preferred:**

Hazen-William's formula: $v = 0.85\, C\, r_H^{0.63} S^{0.54}$

Manning's formula: $v = \frac{1}{n}\, r_H^{2/3} S^{1/2}$

where, v= velocity, m/s; S= slope; r_H= hydraulic radius(=A/P), m;
n = Manning's roughness coefficient; C= Hazen-William's coefficient,

Darcy-Weisbach formula (for pipe flow): $h_L = (fLv^2)/(2gd)$

where, h_L= lead loss, m; d =pipe dia, m; v =velocity, m/s;
L= pipe length, m; f = friction factor (=$4f'$), dimensionless
f'= coefficient of friction g = acceleration due to gravity, m/s^2

Darcy-Weisbach formula

$$H_L = \frac{f'.L.V^2}{2gd}$$

H_L = Head loss in metres

L = Length of pipes in metres

d= Diameter of the pipe in metres

V = Mean velocity of flow through pipe in m/s

g = Acceleration due to gravity

f’ = Dimensionless friction factor generally varying between 0.02 (for smooth new pipes) and 0.075 (for old rough pipes).

Roughness Coefficients and Friction Factors (CPHEEO Manual)

Friction Factor for Darcy-Weisbach formula

Sl. No	Pipe Material	Diameter(mm)		Friction Factor	
		From	To	New	For Design Period of 30 Years
1.	R.C.C.	100	2000	0.01 to 0.02	0.01 to 0.02
2.	A.C	100	600	0.01 to 0.02	0.01 to 0.02
3.	HDPE/PVC	20	100	0.01 to 0.02	0.01 to 0.02
4.	SGSW	100	600	0.01 to 0.02	0.01 to 0.02
5.	C.I. (for corrosive waters)	100	1000	0.01 to 0.02	0.053 to 0.03
6.	C.I (for non-corrosive waters)	100	1000	0.01 to 0.02	0.034 to 0.07
7.	Cement Mortar or Epoxy Lined metallic pipes (Cast Iron, Ductile Iron, Steel)	100	2000	0.01 to 0.02	0.01 to 0.02
8.	G.I.	15	100	0. 014 to 0.03	0.0315 to 0.06

- The dimensionless friction factor depends upon Reynold number and relative roughness of the pipe.
- Relative roughness of the pipe depends upon the absolute roughness (e) of the inside surface and the diameter of the pipe(d).
- Reynold number, $R_e = \frac{Vd}{v}$
- Relative roughness, $\delta = \frac{2e}{d}$

- The approximate values of f’ are given by the following empirical formula.

$$f' = 0.04\left[1 + \frac{1}{35d}\right] \text{ for old pipes}$$

$$f' = 0.02\left[1 + \frac{1}{35d}\right] \text{ for new pipes}$$

- The accurate value of f’ depends upon R_e and δ and given by a formula,

$$f' = a + \frac{b}{R_e{}^m}$$

Where a,b,m are constants depending upon δ.

a) For laminar flows, R_e = 2000

$$f' = \frac{64}{R_e}$$

b) Above R_e = 4000, turbulent flow is fully established.

– Relation suggested by Nikuradse and others

$$\left[\frac{1}{\sqrt{f'}} = 2log_{10}\, Re\, \sqrt{f'} - 0.8\right] \text{ for smooth pipes}$$

$$\left[\frac{1}{\sqrt{f'}} = 2log_{10}\, \frac{d}{2e} + 1.74\right] \text{ for rough pipes}$$

– For smooth pipes and R_e = 20000 to 2000000, slitcher has given the formula

$$f' = 0.005 + \frac{0.396}{R_e{}^{0.3}}$$

c) Nickuradse has given for smooth pipes and for R_e varying between 20000 to 32,40,000 as

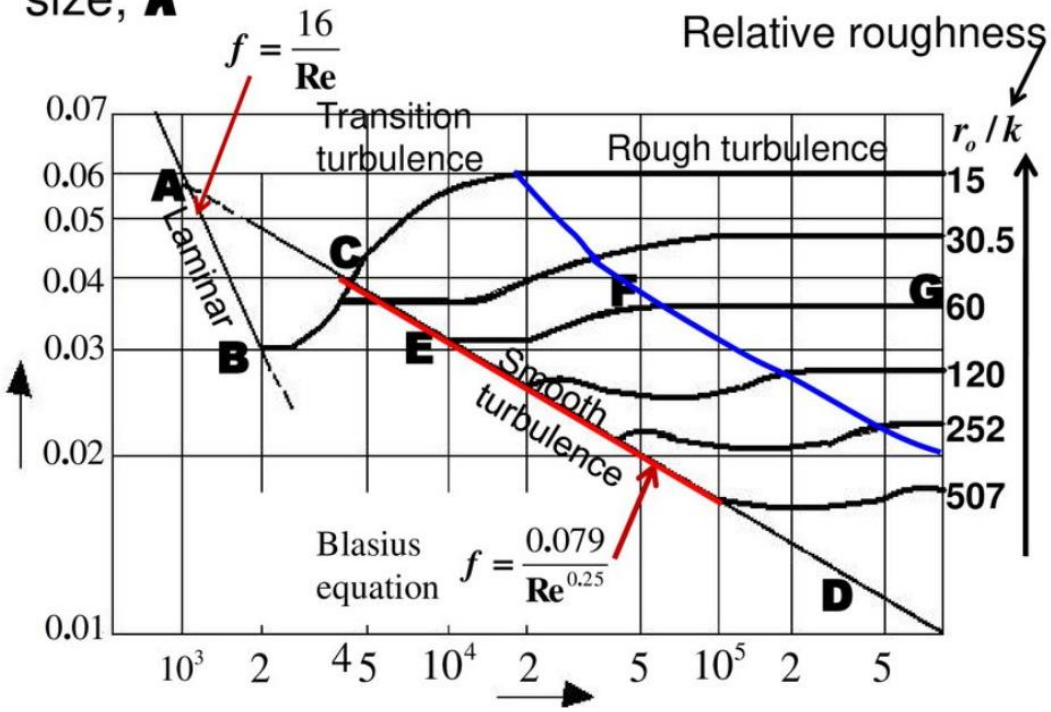
$$f' = 0.0032 + \frac{0.221}{R_e^{0.237}}$$

In the critical zone between R_e = 2000 and R_e = 4000, the relation suggested by Colebrook and White is,

$$\frac{1}{\sqrt{f'}} = 1.74 - 2\,log10\left[\frac{2e}{d} + \frac{18.7}{R_e\sqrt{f'}}\right]$$

Nikuradse's Experiment

- Nikuradse made great progress in 1930's
- Artificially roughened pipes with sand of known size, ***k***



The following points must be noted from the curve

- The line ***AB*** is common to all the pipes having different relative roughness. This line indicates laminar flow. The friction factor in this range is given as

$$f = \frac{16}{Re}$$

Valid for $Re < 2000$

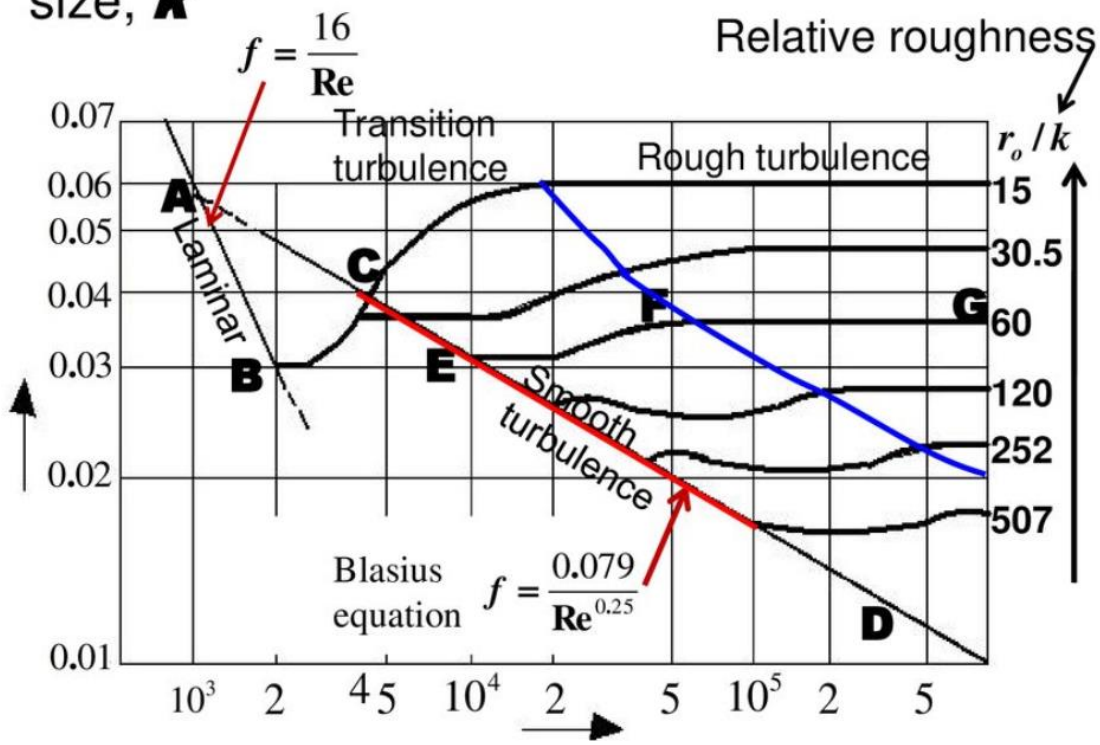
As the value **Re** increases beyond 2000, the flow passes through a transition stage represented by the curve ***BC***.

- The flow becomes turbulent at point ***C***.
- Transition stage occurs in the range of **Re** between 2000 to 4000.

Nikuradse's Experiment

Nikuradse made great progress in 1930's

Artificially roughened pipes with sand of known size, ***k***



The curve ***CD*** represents smooth turbulent flow

For turbulent flow, it is observed that the higher the ratio r_o/k (the smoother the pipe), the greater is the tendency of the pipe to follow the line ***CD***.

- Meaning the pipe with the roughness surface causes the earliest breakaway from the line ***CD***.

There is a **Transition stage** between the **smooth turbulent** flow to the **rough turbulent** flow.

- For example, the relative roughness $r_o/k = 60$, the transition stage is represented by the curve ***EF***.
- In the transition stage, f depends on both **Re** and r_o/k.

After the flow is completely established as rough turbulent, the curves become horizontal.

- For example, the relative roughness $r_o/k = 60$, it is represented by ***FG***.
- In this stage, f depends only on the relative roughness and is independent of **Re**.

Values for 2*e*

S.No	Material of pipe	Value of 2e in mm x 10^{-3}
1	Glass, copper, lead, or asbestos	1.524
2	Steel or wrought iron	45.75
3	Asphalted cast iron	122
4	Galvanized iron	152
5	Cast iron	300
6	Concrete	300 – 3000
7	Riveted steel	900 - 9000

Limitations of Darcy-Weisbach formula

- Very old
- Because of uncertain values of 2e and f', not generally used by water engineers. Even e is known, d is not pre-known
- A trial and error solution is required and making it complicated.

Manning's formula

- Used for gravity conduits
- Also applicable to turbulent flow in pressure conduits
- Yields good result provided the roughness coefficient *n* is accurately estimated.
- Head loss, is given by

$$H_L = \frac{n^2 . V^2 . L}{R^{4/3}}$$

where, n = Manning rugosity coefficient

L = Length of pipe in meters

V = Flow velocity through pipe in m/s

R = Hydraulic mean depth of pipe

$$= \left[\frac{A}{P} = \frac{\frac{\pi d^2}{4}}{\pi d} = \frac{d}{4}\right] \; in\ metres.$$

Roughness Coefficients and Friction Factors (CPHEEO Manual)

Manning's Roughness Coefficient

Type of lining	Condition	n
Glazed coating of enamel Timber	In perfect order	0.010
	(a) Plane boards carefully laid	0.014
	(b) Plane Boards inferior workmanship or aged,	0.016
	(c) Non-plane boards carefully laid	0.016
	(d) Non-plane boards inferior workmanship or aged	0.018
Masonry	(a) Neat cement plaster	0.013
	(b) Sand and cement plaster	0.015
	(c) Concrete, Steel troweled	0.014
	(d) Concrete, wood troweled	0.015
	(e) Brick in good condition	0.015
	(f) Brick in rough condition	0.017
	(g) Masonry in bad condition	0.020
Stone work	(a) Smooth, dressed ashlar	0.015
	(b) Rubble set in cement	0.017
	(c) Fine, well packed gravel	0.020

Type of lining	Condition	n
Earth	(a) Regular surface in good condition	0.020
	(b) In ordinary condition	0.025
	(c) With stones and weeds	0.030
	(d) In poor condition	0.035
	(e) Partially obstructed with debris or weeds	0.050
Steel	(a) Welded	0.013
	(b) Riveted	0.017
	(c) Slightly tuberculated	0.020
	(d) Cement Mortar lined	0.011
Cast Iron & Ductile Iron	(a) Unlined	0.013
	(b) Cement mortar lined	0.011
Asbestos Cement		0.012
Plastic (smooth)		0.011

HAZEN WILLIAM'S FORMULA

$$V = 0.85\, CH . R^{0.63} . S^{0.54}$$

Where,

CH = Coefficient of hydraulic capacity

R = Hydraulic mean depth of pipe in metres

$= \left(\frac{d}{4}\right)$ for circular pipes flowing full

S = Slope of energy line

V = Flow velocity through the pipe in m/s.

Pipe material	Value of C_H Depending upon the smoothness of the pipe material
Concrete(regardless of age)	130
Cast Iron	
New	130
5 years old	120
20 years old	100
Welded steel (new)	120
Riveted steel (new)	110
Vitrified clay	110
Brick sewers	100
Asbestos-Cement	140

Roughness Coefficients and Friction Factors (CPHEEO Manual)

Hazen-William's Constant

Pipe Material	Recommended C Values	
	New Pipes[@]	Design Purpose
Unlined Metallic Pipes		
Cast Iron, Ductile Iron	130	100
Mild Steel	140	100
Galvanized Iron above 50 mm dia. #	120	100
Galvanized Iron 50 mm dia and below used for house service connections. #	120	55
Centrifugally Lined Metallic Pipes		
Cast Iron, Ductile Iron and Mild Steel Pipes lined with cement mortar or Epoxy		
Up to 1200 mm dia	140	140
Above 1200 mm dia	145	145

Pipe Material	Recommended C Values	
	New Pipes[@]	Design Purpose
Projection Method Cement Mortar Lined Metallic Pipes		
Cast Iron, Ductile Iron and Mild Steel Pipes	130*	110**
Non Metallic Pipes		
RCC Spun Concrete,		
Prestressed Concrete		
Up to 1200 mm dia	140	140
Above 1200 mm dia	145	145
Asbestos Cement	150	140
PVC, GRP and other Plastic pipes.	150	145

Pumps for Water Withdrawal and Conveyance

- Pumps are devices which convert mechanical energy into hydraulic energy for lifting the water or any fluid to higher elevations or at higher pressure. Pumps are employed in water supply projects at various stages for different purposes:



- To lift raw water from wells to the treatment works.
- For filling elevated distribution reservoirs or overhead tanks.
- To deliver treated water to the consumer at desired pressure.
- To supply pressured water for fire hydrants.
- To boost up pressure in water mains.
- To fill elevated overhead water tanks.
- To back-wash filters.
- To pump chemical solutions, needed for water treatment.

Classification of Pumps

 Pumps are mainly divided into two major categories, *Dynamic (or Kinetic) Pumps* and *Displacement (or Positive Displacement) Pumps*. However, there are various sub-classifications,

Based on principle of operation:

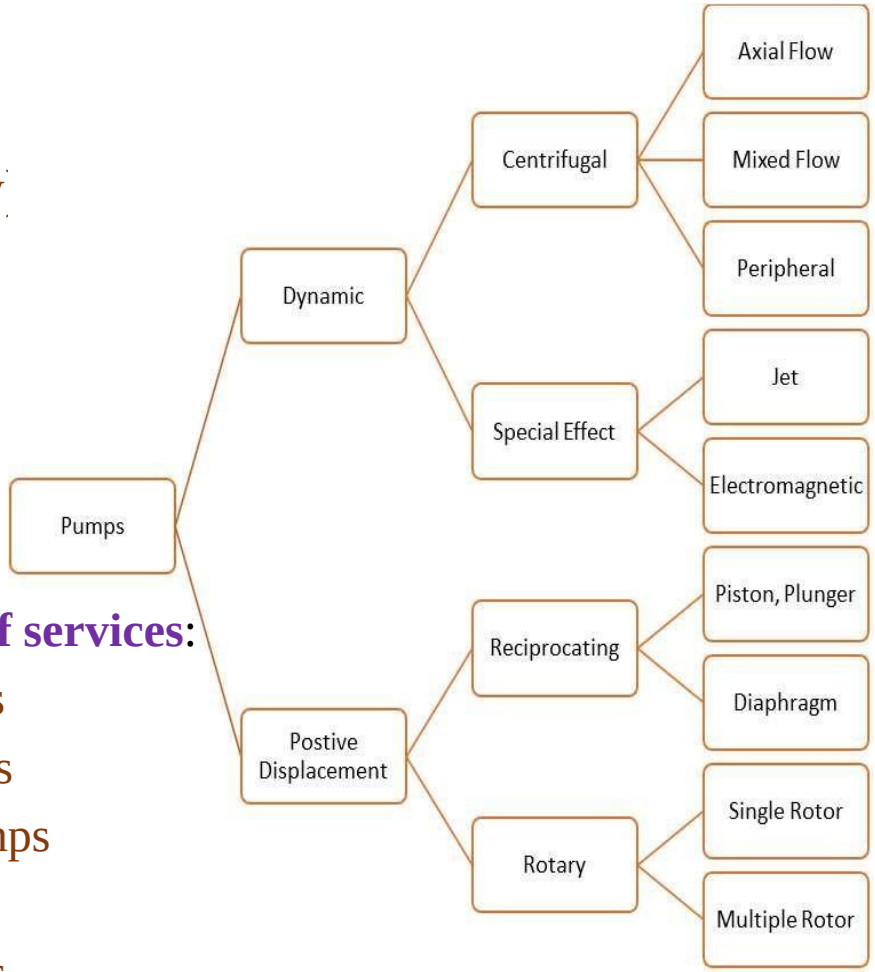
- Displacement pumps (reciprocating, rotary)
- Velocity pumps (centrifugal, turbine, jet)
- Buoyancy pumps (air lift pumps)
- Impulse pumps (hydraulic rams)

Based on type of power required:

- Electrical driven pumps
- Gasoline engine
- Steam engine pumps
- Diesel engine pumps

Based on type of services:

- Low lift pumps
- High lift pumps
- Deep-well pumps
- Booster pumps
- Standby pumps

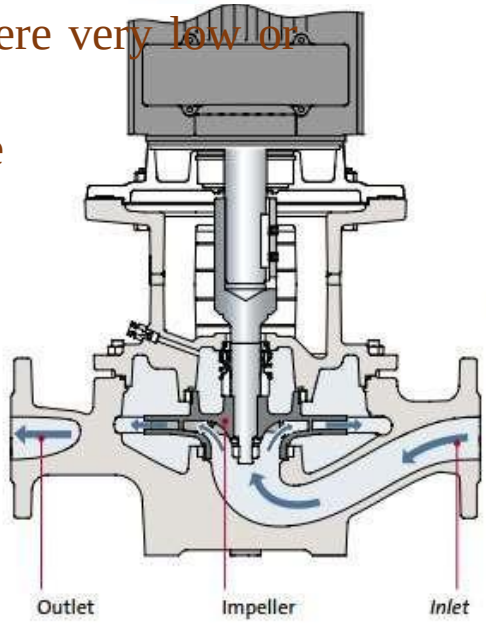


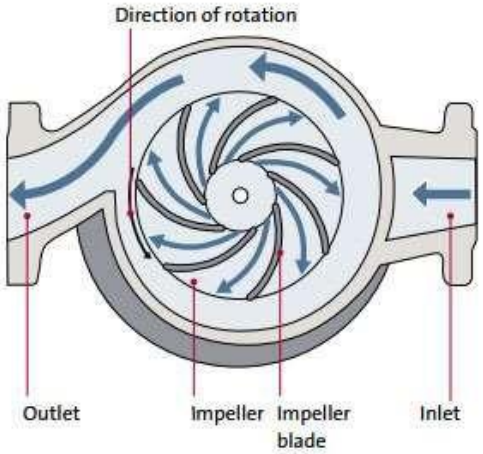
Centrifugal Pumps

- The water which enters inside the pump is revolved at high speed by means of impeller and is thrown to the periphery by the centrifugal force.
- Centrifugal pumps are the most used pump worldwide, due to simple working principle and relatively inexpensive manufacturing cost.
- Due to compact design, they require very small space, are easy to move and provide a lot of flexibility.
- They have rotary motion due to which there very low or no noise
- They are not damaged due to high pressure







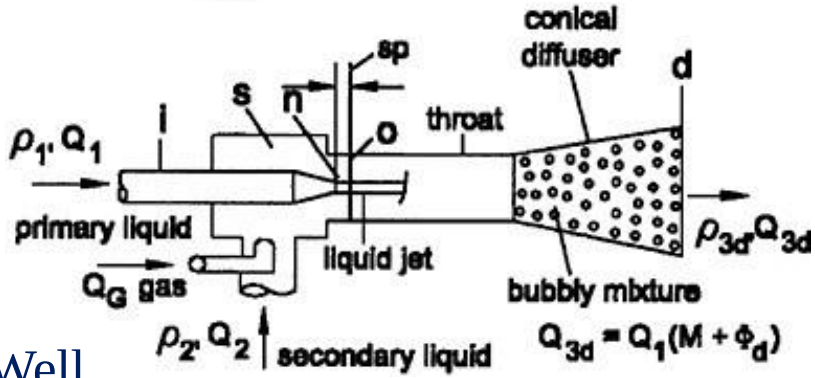


- The main disadvantage is that they have almost no suction power (use rotation insteed), and thus must be put under water, or primed, before it will move water.
- Centrifugal pumps can also develop “cavitation” when the speed of the water causes it to vaporise, creating

Jet Pumps

- Jet pumps, powered by an electric motor driving an impeller, moves water from the well through a narrow orifice, or jet, mounted in the housing in front of the impeller. This constriction at the jet causes the speed of the moving water to increase, and as the water leaves the jet, a partial vacuum is created that sucks additional water from the well. Directly behind the jet is a Venturi tube that increases in diameter to slow down the water and increase the pressure.





- Different types available: Shallow Well, Deep Well, Submersible.

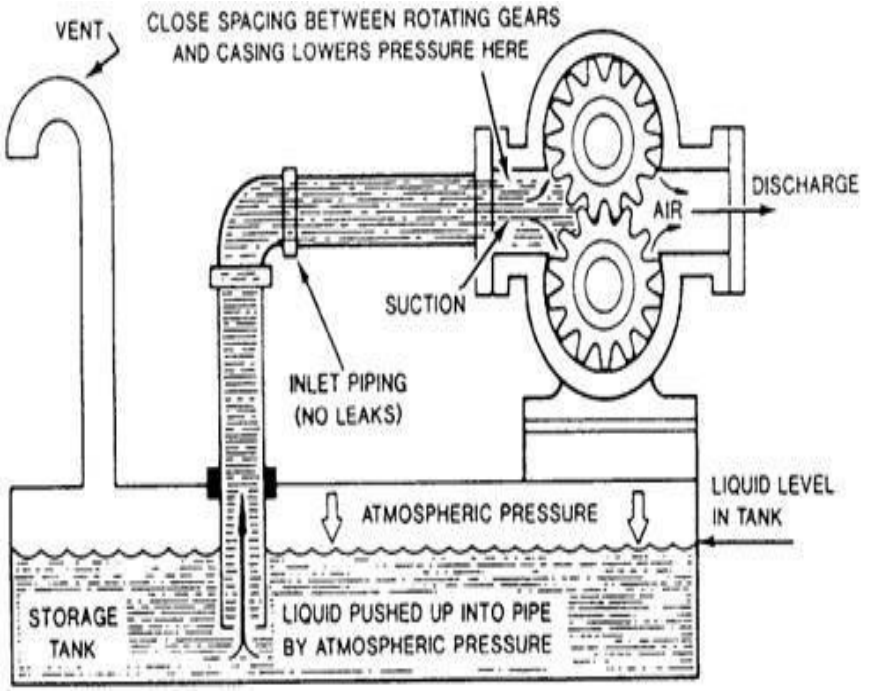
- These days, these are most common pump for a shallow well.
- Depending on type, can be submersible or offsets from the well.
- Inexpensive and require less maintenance.
- Adaptable to different depths and can pump water from greater depth
- Its efficiency reduces as the total suction lift increase.
- Gets damaged easily by sand.

Rotary Pumps

- These are positive displacement type pumps which create vacuum mechanically by the movable part of the pumps, where for each revolution of the pump, a fixed volume of fluid is moved regardless of the resistance against which the pump is pushing.

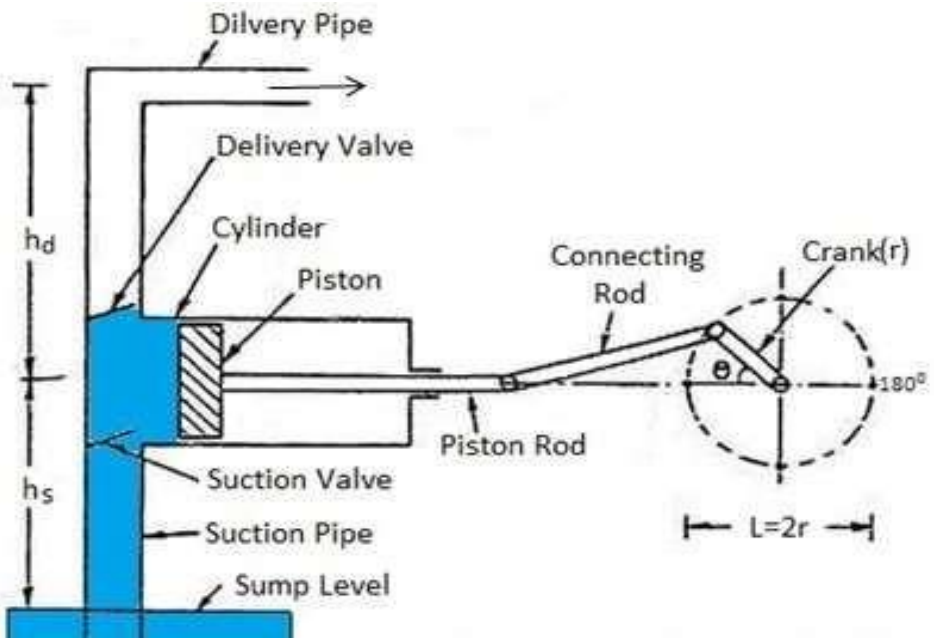


- Examples are gear pump, screw pump, and moving vane pump.
- Rotary pumps are easy in construction and maintenance, and have no valves.
- Gives steady and constant flow at high efficiency, especially at low to moderate heads.
- Do not require any priming as they are self-primed
- Often used for the individual building water supply and for fire protection
- High initial cost as well as maintenance cost due to gears abrasion
- Cannot pump water containing suspended impurities as the wear and abrasion caused will destroy the seal between the cans and the casing, also any blockage in the system could quickly cause damage to the pump or a



Reciprocating Pumps

- Reciprocating pump is a positive displacement pump operating on the principle of pushing of liquid by a piston that executes a reciprocating motion in a closed fitting cylinder. During suction stroke, it sucks liquid and during discharge or return stroke, delivers liquid under pressure. The discharge depends on the swept volume and the pump speed.
- Examples are Piston pump, Plunger pump, and Diap



- Reciprocating pumps operate at higher efficiency compared to other pump. In most cases, at any set point, reciprocating pumps operate around 90%.
- Gives high suction lift and high pressure at outlet.
- Do not require any priming
- suitable for low volumes of flow at high pressures



- The main disadvantage of a reciprocating pump is the high maintenance and short life, due to high wear and tear.
- Flow is less and non-uniform, thus cannot be used for high flow operations.
- More heavy and bulky in shape.

Capacity of Pumps

- ***Total head (H, m) to be supplied by pump***

  = ***Suction head (h_s) + delivery head (h_d) + friction loss (h_f)*** + Minor losses (due to exit, entrance, bends, valves etc.)

- ***Work done by the pump***

  = mass of water to be lifted (m, kg). acceleration due to gravity (g, m/s^2). total head to be covered (H, m)

  = mgH = ρVgH [as m = ρV, where ρ is density of water, kg/m^3 and V volume of water pumped, m^3]

- ***Power Required (P, Watt)***

  = Work performed per unit time (t, s) = ρVgH/t = (ρg)(V/t)H ***= γQH***

  [as V/t = Q, and ρg = γ, where γ is specific weight of water, N/m3 and Q is discharge of pump, m^3/s]

Capacity of Pumps

- *Hydraulic Horse Power or Water Horse Power (HP)* *[1 HP = 746 W = 0.746 kW]*

  = γQH/746 *[γ_{water} = 9807 N/m³ = 9.807 kN/m³]*

- *Break Horse Power (BHP) required, considering η as overall efficiency*

  =HP/η = (γQH/746)/η

The total BHP required may be supplied with a number of pumps. As a common practice, even number of pumps say 2,4,... are provide and half of the pumps are required as stand-by.